

DermaXplain: Evaluating Explainable AI Methods Against Clinically Structured Pseudo-Concepts in Skin Lesion Classification

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Abstract—Convolutional neural networks can achieve strong performance in dermoscopic skin-lesion classification, but their decision processes remain opaque and difficult to verify clinically. Post-hoc explainable AI methods (XAI) are increasingly used to interpret such models, but standard evaluation often measures only whether saliency falls inside the lesion boundary. This provides limited evidence of clinical relevance, since dermatological assessment depends on structured visual cues such as asymmetry, border irregularity, and colour heterogeneity. This paper presents DermaXplain, a framework that evaluates Grad-CAM, SHAP, and LIME against automatically generated ABCD-inspired pseudo-concept maps. These spatial maps allow direct comparison with saliency outputs. Experiments on HAM10000 and ISIC 2018 showed that lesion-level localisation was much stronger than pseudo-concept alignment. SHAP achieved the highest raw overlap, while colour heterogeneity showed the most consistent concept-level alignment. However, chance-adjusted evaluation changed the ranking: Grad-CAM showed stronger enrichment for asymmetry and border irregularity, while part of SHAP’s raw advantage was explained by geometry and saliency coverage. The results show that XAI evaluation in dermoscopy should combine lesion localisation, pseudo-concept alignment, and random-baseline correction.

Index Terms—Dermoscopy, Explainable AI, Melanoma Classification, Pseudo-Concepts, Saliency Mapping, ResNet-50.

I. INTRODUCTION

Early detection of melanoma remains the most critical factor influencing patient survival rates, establishing dermoscopy (also called dermatoscopy) as an indispensable non-invasive skin imaging technique. While modern Convolutional Neural Networks (CNNs) have achieved expert-level classification performance on vast public medical image banks, their translation into live diagnostic environments is severely throttled by their opaque, “black-box” representation layers. Consequently, there is an urgent and critical need for trustworthy, clinically accepted Explainable Artificial Intelligence (XAI) frameworks that can be quantitatively validated. Post-hoc Explainable AI methods attempt to address this opacity by producing spatial explanations that indicate which image regions contribute to a model prediction. However, evaluating whether these explanations are clinically meaningful remains difficult. The core motivation of this study is that traditional lesion-level

overlap assessments are fundamentally insufficient for clinical verification. Professional dermatologists do not inspect a lesion as a flat, homogeneous block of pixels; instead, they reason systematically using highly structured visual sub-features. This study evaluates three foundational post-hoc explainability approaches: Grad-CAM, SHAP, and LIME by introducing a middle evaluation layer composed of algorithmically extracted, ABCD-inspired pseudo-concept maps. We fundamentally re-frame the XAI validation problem to explore: *How reliable are current XAI evaluation approaches under varying lesion conditions and pseudo-concept structures?* This approach, demonstrates why evaluation itself becomes unstable due to sparse concept structures, shrinking lesion scales, high metric sensitivity, and clear method disagreements. The evaluation is carried out using four complementary metrics to capture various aspects of explanation localisation: IOU (Intersection over Union), Dice coefficient, Saliency Inside Ratio (SIR) and Trustworthiness Index for AI (TIxAI).

To guide this structural analysis, we formulate and address the following targeted research questions:

RQ1: How well do Grad-CAM, SHAP, and LIME localise saliency within lesion regions?

RQ2: To what extent do XAI explanations align with clinically inspired pseudo-concept regions derived from lesion masks?

RQ2a: To what extent does lesion size influence pseudo-concept alignment and the reliability of XAI evaluation metrics?

RQ2b: For diagnostically labelled HAM10000 images, do pseudo-concept alignments differ between melanoma and non-melanoma lesions?

RQ2c: Do XAI methods align with pseudo-concept regions more strongly than expected from random overlap within the lesion?

Collectively, these questions investigate whether current saliency-based XAI methods identify clinically meaningful dermoscopic pseudo-concepts, or whether apparent agreement is explained by lesion geometry, diagnostic composition, and chance spatial overlap within lesions.

Following the introduction, the literature review establishes

the clinical background, relevant dermoscopic images, deep learning approaches, saliency-based XAI methods, and current limitations in lesion and concept-level explanation evaluation. The shared dataset model, XAI, and evaluation definitions are then presented, after which the remainder of the report is organised sequentially by research question. For each question the relevant motivation, methodology, results, are presented together.

II. LITERATURE REVIEW

A. Clinical Background and Diagnostic Criteria

Dermoscopy uses direct lighting and magnification to reveal structures that are not visible to the naked eye. Clinical assessment of pigmented lesions relies on localised visual patterns involving colour, texture, and lesion geometry. However, dermoscopic interpretation requires specialist training and remains subject to observer variability [1].

To improve diagnostic consistency, clinical practice has historically relied on structured heuristic frameworks, most notably the classic ABCD rule popularised by Nachbar et al. [2]. This established clinical protocol scores and stratifies high-risk melanocytic skin lesions based on four distinct visual axes: Asymmetry (A), Border irregularity (B), Colour variation (C), and Diameter (D) [2]. Early computer-aided diagnosis systems attempted to operationalise these criteria using handcrafted descriptors of lesion shape, border structure, and colour distribution [3]. Recent machine learning studies continue to use these clinical criteria, through explicit ABCD-based feature extraction or through hybrid AI-supported diagnostic workflows [4], [5]. These clinical criteria motivate the pseudo-concepts used in this work.

B. Deep Learning and CNNs in Dermatology

Deep learning shifted dermoscopic image analysis from handcrafted feature engineering to end-to-end convolutional neural networks trained directly on image pixels. Public datasets such as ISIC 2018 and HAM10000 have supported this progress by providing large collections of dermoscopic images with diagnostic labels and lesion masks [1], [6]. CNN-based models can achieve strong classification performance, but their internal representations are difficult for clinicians to inspect or verify. As Litjens et al. observe, this emphasis on predictive performance has reduced the explicit modelling of clinically interpretable concepts, leaving modern end-to-end deep learning systems opaque to medical practitioners [7].

C. Saliency-Based XAI Methods

Post-hoc XAI methods are commonly used to generate visual explanations of deep neural network predictions for clinical interpretation [8].

A prominent and widely implemented approach within medical imaging and dermatological classification pipelines is Gradient-weighted Class Activation Mapping (Grad-CAM), introduced by Selvaraju et al. [9]. Grad-CAM operates by capturing the class-specific gradient flows backpropagated into the final convolutional layer of a trained network. By

computing a rectified linear unit (ReLU) over the coarse combination of these forward activation maps, it produces a spatial heatmap highlighting the macroscopic regions that most heavily influenced a given prediction [9]. In dermoscopy, this reveals whether a classifier aligns with the skin lesion or focuses on irrelevant confounding artifacts (e.g., dark hair or gel bubbles).

Concurrently, model-agnostic feature attribution techniques are deployed at a single-instance level to probe black-box models without modifying their internal architectures. Local Interpretable Model-agnostic Explanations (LIME), formulated by Ribeiro et al., isolates an individual input image and approximates the localised decision boundary of a complex model [10]. It accomplishes this by fitting an interpretable sparse linear surrogate model over randomly perturbed super-pixel segmentation patches, explicitly assigning weights that denote which local image zones contributed to or detracted from the target prediction [10].

Alternatively, Shapley Additive exPlanations (SHAP) builds upon game-theoretic principles to enforce essential mathematical axioms, such as local accuracy, consistency, and missingness [11]. As detailed by Srivastava et al., SHAP has become an indispensable tool across high-stakes domains, including medical diagnostics and financial markets. It computes additive feature importance scores that quantify explicit pixel-level contributions [11].

When deployed in modern dermatological diagnostic frameworks, these three post-hoc methods show distinct behavioural differences and visual contradictions. As demonstrated in the explainable skin frameworks constructed by Nigar et al. [12], Mridha et al. [13], and Metta et al. [14], these tools provide complementary but highly discordant visual fields. Grad-CAM generates coarse, sweeping, and spatially continuous activation fields indicating general regional alignment [9]. In contrast, instance-level techniques like LIME and SHAP offer highly granular, diffuse, or fragmented pixel-level attributions that attempt to map fine-grained feature boundaries [10]. This behavioural fluctuation leaves clinician verification vulnerable to subjective interpretation, creating a clear need to quantitatively structure and validate these contradictory attributions.

D. Evaluation Techniques and Concept-Based XAI

Evaluating the reliability and clinical relevance of post-hoc explainability maps in medical imaging remains an important open challenge [8], [11]. Historically, many deep diagnostic pipelines have relied entirely on ad-hoc or purely qualitative human visual inspection. Dhar et al. heavily criticises such approaches as unscientific, subjective, and highly dangerous for safety-critical healthcare fields [8]. To establish objective benchmarks, localisation-aware pipelines developed by Nigar et al. [12] and Mridha et al. [13] have transitioned towards mask-based spatial evaluation strategies. These approaches leverage expert-annotated lesion segmentation masks, such as those provided within the ISIC 2018 [1] and HAM10000 [6] datasets, as spatial reference regions for evaluating explanation localisation. Classical segmentation metrics are adapted for

explanation evaluation: Intersection over Union (IoU) and the Dice coefficient quantify the geometric overlap between binarised saliency maps and reference lesion masks. In contrast, the Saliency-Inside Ratio (SIR) provides an intensity-based concentration measure by calculating the proportion of total saliency mass located within the lesion boundary rather than in the surrounding background or image artefacts.

To address broader dimensions of clinical trust and regional alignment, Ieracitano et al. proposed TlxAI, a multidimensional Trustworthiness Index for skin lesion classification that mathematically formalises and weights the differences in feature relevance and consistency between lesion and non-lesion zones [15].

To move beyond whole-lesion overlap, recent approaches have introduced more structured evaluation that relate explanations to clinically meaningful lesion characteristics [16].

E. Positioning the Research Gap

Despite progress in explainable medical imaging, existing approaches still face important limitations. Post-hoc explanations are often assessed visually or evaluated against whole lesion masks. This indicates whether saliency falls within the lesion but not whether it corresponds to specific clinically inspired structures [8], [12], [13], [15]. Existing dermatological image analysis methods have also quantified ABCD-related properties such as asymmetry, border irregularity, and colour variation using handcrafted measurements or scalar indices [3], [16], [17]. These measures can describe the overall severity of a characteristic. They do not, however, identify the spatial regions responsible for that measurement and therefore cannot be directly compared with saliency maps.

The literature therefore reveals a gap between lesion-level explanation evaluation and the structured visual cues used in dermoscopic assessment. Metrics such as IoU, Dice, and SIR treat the complete lesion mask as a single reference region. They do not distinguish whether saliency aligns with asymmetry, border irregularity or colour variation. A strong lesion localisation result does not necessarily indicate alignment with clinically inspired internal structures. Furthermore, these alignments may be affected by pseudo-concept sparsity, lesion size or just correspond to expected random spatial overlap.

DermaXplain addresses this gap by generating ABCD-inspired pseudo-concept maps rather than single numerical indices. These maps provide spatially localised proxies for asymmetry, border irregularity, and colour heterogeneity, enabling direct comparison with Grad-CAM, SHAP, and LIME saliency outputs. The framework evaluates both raw and chance-adjusted alignment and examines how results vary with lesion size, concept geometry, and diagnostic group. To support this evaluation, the study requires dermoscopic datasets that provide both diagnostic information and lesion-level spatial references.

III. DATASETS AND PREPROCESSING

A. Datasets and Segmentation Masks

Public dermoscopic datasets support these requirements. **HAM10000** provides diagnostic labels for skin-lesion classification, while **ISIC 2018 Task 1**, provides expert-annotated lesion segmentation masks for spatial analysis [1], [6].

Binary lesion masks identify lesion pixels as 1 and background pixels as 0. They were used to define lesion reference regions, derive lesion-size and border-contact metadata, generate pseudo-concept maps, and support lesion-level localisation evaluation. The masks serve as reference annotations for lesion boundaries, but they do not provide ground truth for internal clinical concepts.

B. Preprocessing Pipeline

A unified preprocessing pipeline was applied to both datasets to ensure consistent image-mask pairing and to generate the metadata required for classification and explainability analysis. Inventory manifests produced during data auditing were first filtered to retain only samples with an existing image, an existing mask, and matching image-mask dimensions.

For model input and spatial evaluation, RGB images were resized to 224×224 pixels using bilinear interpolation, while binary masks were resized using nearest-neighbour interpolation to preserve discrete lesion boundaries. RGB inputs were scaled to $[0, 1]$ and normalised using ImageNet channel statistics, consistent with the pre-trained ResNet-50 backbone. The common 224×224 grid was subsequently used for model inputs, saliency maps, and pseudo-concept maps.

Lesion-level morphology descriptors were extracted from the binary masks, including lesion area, relative mask coverage, equivalent diameter, centroid coordinates, and bounding-box dimensions. Border-contact indicators recorded whether the lesion reached the top, bottom, left, or right image boundary and the number of borders touched. Lesions were also grouped into four predefined size classes: Normal, Small, Tiny, and Very Tiny, using the `mask_size_class` metadata employed throughout the subsequent stratified analyses.

The resulting descriptors were stored in enriched preprocessing manifests and reused during model training, pseudo-concept generation, lesion-size analysis, and chance-adjusted evaluation. Following initial visual and pipeline sanity checks, the final explainability experiments were conducted on the complete available datasets.

C. Dataset Morphology Diagnostics

The preprocessing manifests were used to characterise lesion-scale distribution and border contact across the combined HAM10000 and ISIC 2018 evaluation set. Size-class prevalence was summarised over the four predefined scale tiers, while border diagnostics quantified the frequency with which lesions reached one or more image boundaries. Lesion scale and boundary contact were retained as potentially relevant morphological variables. **Table I** reports the lesion-size distribution used in the stratified analysis; border-contact

information was preserved in the preprocessing metadata but was not analysed separately in the main report.

TABLE I: Dataset Structural and Morphological Stratification

Scale Tier	Count	Dataset %
Normal ($\geq 5\%$)	11254	89.3
Small ($2\% - < 5\%$)	971	7.7
Tiny ($1\% - < 2\%$)	285	2.3
Very Tiny ($< 1\%$)	99	0.8

IV. BASELINE CLASSIFICATION MODEL

A. Architecture and Transfer Learning

The baseline classifier utilises a ResNet-50 backbone initialised with pre-trained ImageNet weights. The lower convolutional blocks are locked to preserve generalised feature representations, while the upper residual blocks and classification heads are unfrozen to adapt to localised dermoscopic structures.

B. Training Mechanics

The network is optimised using an Adam optimiser with an initial learning rate of 10^{-4} paired with a cosine annealing schedule over 50 epochs using a batch size of 32. Class imbalance is handled during backpropagation via class-weighted cross-entropy loss tracking.

C. Performance Evaluation

The purpose of the classifier was to provide a sufficiently discriminative and stable prediction model for the subsequent XAI evaluation, rather than establish a new state-of-the-art classification result. Its validation performance was therefore considered adequate for the primary objective of the study: analysing spatial behaviour and evaluation reliability of post-hoc explanations.

The fine-tuned ResNet-50 model achieves stable, high-tier validation performance across all primary classification metrics, which are summarised in **Table II**.

TABLE II: Baseline Classifier Validation Results

Metric	Accuracy	Precision	Recall	Macro F1	AUC-ROC
Value	0.894	0.737	0.787	0.759	0.909

V. EXPLAINABLE AI METHODS

Three complementary post-hoc explanation methods were evaluated: Grad-CAM, SHAP, and LIME. Their theoretical foundations are reviewed in the previous Literature Review, Section II-C; here, they are considered in terms of the spatial explanation patterns generated for the ResNet-50 classifier.

Grad-CAM produces coarse, spatially smooth localisation maps from gradients propagated through the final convolutional layer. SHAP generates additive attribution maps that are typically more diffuse and spatially distributed, while LIME identifies influential superpixel regions through local

perturbation and surrogate modelling. These contrasting behaviours are important for quantitative evaluation because broad, diffuse, and fragmented saliency patterns interact differently with lesion masks and sparse pseudo-concept regions. The three methods are therefore evaluated using the same lesion-level, pseudo-concept, and chance-adjusted protocols. A representative example of the three explanation patterns is shown in **Fig. 1**.

VI. EVALUATION METRICS

Four complementary metrics were used for lesion-level and pseudo-concept evaluation. Let H denote the normalised continuous saliency map generated by an XAI method, X the corresponding binary top- $k\%$ saliency region, and M a binary reference mask representing either the complete lesion or a pseudo-concept region. In this study, X was obtained by retaining the top- $k\%$ of saliency values, with $k = 20\%$.

The metrics capture complementary aspects of explanation localisation. IoU and $Dice$ measure strict spatial overlap between the binary salient region X and the reference mask M . SIR and the normalised $TIxAI$ -style score instead operate on the continuous saliency map H . SIR measures the proportion of total saliency mass located inside the reference region. Referring to Ieracitano et al., [15], $TIxAI$ compares the mean saliency density inside and outside it.

Full pixel-wise definitions are provided in the Appendix ¹.

$$IoU(X, M) = \frac{|X \cap M|}{|X \cup M|} \quad (1)$$

$$Dice(X, M) = \frac{2|X \cap M|}{|X| + |M|} \quad (2)$$

$$SIR(H, M) = \frac{\sum_{i,j} H_{i,j} M_{i,j}}{\sum_{i,j} H_{i,j}} \quad (3)$$

$$TIxAI_{norm}(H, M) = \frac{RD_{in}}{RD_{in} + RD_{out} + \epsilon} \quad (4)$$

where

$$RD_{in} = \frac{\sum_{i,j} H_{i,j} M_{i,j}}{\sum_{i,j} M_{i,j}} \quad (5)$$

and

$$RD_{out} = \frac{\sum_{i,j} H_{i,j} (1 - M_{i,j})}{\sum_{i,j} (1 - M_{i,j})} \quad (6)$$

These four metrics are used throughout the analyses for RQ1, RQ2, RQ2a, RQ2b, and RQ2c, subject to the metric-specific use of binary or continuous saliency representations.

The following sections address the research questions sequentially, presenting the specific analytical problem, method, and results associated with each question.

¹See the supplementary appendix: Section 2

VII. RESEARCH QUESTION 1: LESION-LEVEL SALIENCY LOCALISATION

To what extent do Grad-CAM, SHAP, and LIME localise saliency within lesion regions?

This analysis establishes a lesion-level localisation baseline before the pseudo-concept evaluation in Section VIII. Lesion-level localisation assesses whether XAI explanations concentrate on the lesion region, but should not be interpreted as image segmentation: saliency methods highlight discriminative regions rather than reproduce the complete lesion boundary. Consequently, moderate overlap scores may still reflect meaningful model attention, while strong lesion localisation does not necessarily imply alignment with specific clinically inspired structures such as asymmetry, border irregularity, or colour heterogeneity.

As illustrated in **Fig. 1** the three methods produce different spatial explanation patterns. The example illustrates their distinct spatial behaviours: Grad-CAM produces a coarse localised activation pattern, SHAP distributes relevance more continuously across the lesion, and LIME highlights discrete superpixel regions. Additional qualitative examples are provided in Appendix ².

The lesion-level results are reported in **Table III**. IoU and Dice quantify overlap between the binary top- k % saliency region and the corresponding binary lesion mask, while SIR and the normalised TIxAI-style score evaluate the concentration of continuous saliency within and relative to the lesion region.

TABLE III: Lesion-Level Mask Overlap Summary

Method	IoU	Dice	SIR	TIxAI	Area %
SHAP	0.312	0.461	0.434	0.732	20.0%
LIME	0.257	0.391	0.391	0.674	32.3%
Grad-CAM	0.241	0.358	0.320	0.392	69.4%

SHAP achieved the highest localisation performance across all four evaluation metrics, followed by LIME and Grad-CAM. The largest differences were observed for TIxAI, indicating that SHAP produced the most concentrated saliency within lesion boundaries. Despite generating the largest saliency regions, Grad-CAM obtained the lowest

localisation scores, demonstrating that accurate localisation depends more on saliency placement than overall coverage. These results confirm that all three XAI methods successfully localise lesion regions, with SHAP providing the strongest lesion-level localisation. Section VIII investigates whether this localisation also corresponds to clinically significant dermoscopic concepts.

VIII. PSEUDO-CONCEPT-LEVEL EVALUATION

A. Methodological Motivation

Whole-lesion localisation does not reveal whether an explanation aligns with the more structured visual cues used in dermoscopic assessment. As discussed in Section II, the ABCD rule considers asymmetry, border irregularity, colour variation, and diameter [2] [4].

However, expert spatial annotations for these concepts are rarely available. To bridge this gap, *DermaXplain* introduces algorithmically derived pseudo-concept maps as scalable spatial proxies for clinically inspired features. Unlike prior approaches that summarise properties such as asymmetry using scalar indices, the proposed framework produces spatial maps identifying the regions contributing to each concept. This enables direct comparison with XAI saliency maps. The pseudo-concepts are heuristic evaluation references rather than clinical ground truth.

B. Pseudo-Concept Generation

Pseudo-concept maps were generated on the common evaluation grid used by the ResNet-50 classifier and XAI outputs. This is to ensure direct spatial comparability between saliency and reference maps. Images were resized using bilinear interpolation, while binary lesion masks were resized using nearest-neighbour interpolation to preserve discrete boundaries. Geometric and image-based operations were then applied to derive asymmetry, border-irregularity, and colour-heterogeneity pseudo-concept maps. Diameter was excluded because only few images carry physical reference scale, making absolute size meaningless after resizing. Each image produced a single compressed archive, retaining the lesion mask for traceability and later evaluated against the Grad-CAM, SHAP, and LIME saliency maps.

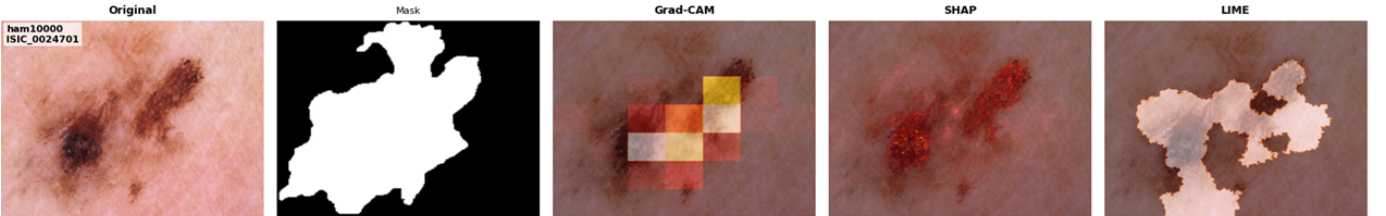


Fig. 1: Representative comparison of Grad-CAM, SHAP, and LIME explanations for a dermoscopic image. The corresponding binary lesion mask is shown for reference.

²See the supplementary appendix: Section 1.1

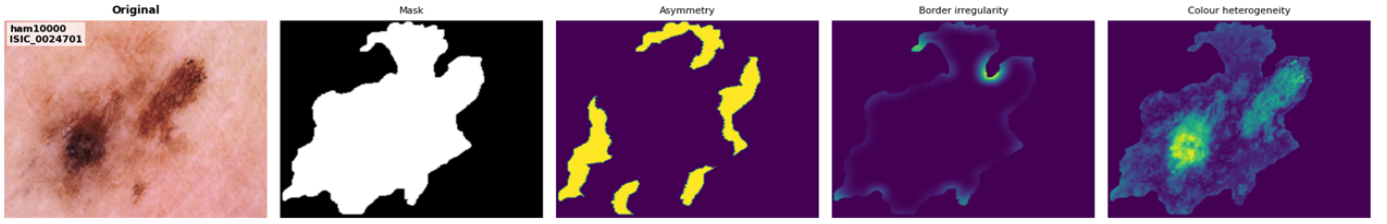


Fig. 2: Representative generation of asymmetry, border-irregularity, and colour-heterogeneity pseudo-concept maps from dermoscopic images and lesion masks.

C. Asymmetry

Following the geometric principles of Ali et al. [16], the lesion mask was analysed using PCA-derived major and minor axes. The mask was reflected across each axis and spatial disagreement between the original and reflected shapes was computed. Major and minor-axis disagreement maps were combined using a pixel-wise maximum operation. Unlike conventional numerical asymmetry indices, this produces a spatial map highlighting the regions responsible for bilateral shape disagreement.

D. Border Irregularity

Border irregularity was represented as a continuous spatial map derived from deviations between the original lesion contour and a smoothed reference contour, following the general motivation of contour-based irregularity analysis [17]. A Gaussian-smoothed lesion mask provided an ideal reference shape, and boundary deviations were propagated into an inner border band. Several parameterisations were evaluated; because they produced comparable alignment behaviour, the distance-weighted formulation was retained for subsequent analysis.

E. Colour heterogeneity

Colour heterogeneity was treated as an image-derived proxy rather than a clinically validated concept annotation. Lesion pixels were transformed to CIELAB space, and the median lesion colour was used as the reference.[16] Each pixel received a value proportional to its Euclidean colour distance from this median, producing a continuous map of relative colour deviation. Because no expert concept-level annotations were available, this map should be interpreted as a heuristic proxy. Unlike asymmetry and border irregularity, colour heterogeneity often covers a larger proportion of the lesion area. Consequently, it is expected to exhibit higher overlap with XAI explanations and provides an important comparison point when interpreting pseudo-concept.

F. Purpose and Characteristics of Pseudo-Concept Maps

The pseudo-concept maps are not intended to reproduce the clinical ABCD assessment process or serve as expert concept annotations. Instead, they provide spatially localised heuristic approximations of selected clinically inspired cues. Asymmetry and border irregularity are derived primarily from

lesion geometry, whereas colour heterogeneity is derived from image appearance. They are therefore used as structured spatial reference maps for evaluating XAI explanations beyond whole-lesion localisation. **Figure 2** illustrates representative pseudo-concept maps used throughout the evaluation. Additional examples are provided in Appendix ³.

Table IV summarises the mean pseudo-concept area as a percentage of lesion area. Diagnostic comparisons are based on HAM10000, while the overall mean combines HAM10000 and ISIC 2018.

TABLE IV: Mean Pseudo-Concept Area as a Percentage of Lesion Area

Concept	Melanoma	Non-Melanoma	Overall
Asymmetry	4.72	3.19	3.20
Border Irregularity	0.60	0.58	0.57
Colour Heterogeneity	5.24	4.11	4.04

Table IV shows that across the combined evaluation set, colour heterogeneity occupied the largest mean proportion of the lesion (4.04%), followed by asymmetry (3.20%), while border irregularity occupied less than 1%. Within HAM10000, melanoma lesions exhibited larger mean asymmetry and colour heterogeneity regions than non-melanoma lesions, while the border-irregularity area differed little between groups. These differences in pseudo-concept size and sparsity are important when interpreting the subsequent overlap and chance-adjusted analyses.

IX. PSEUDO-CONCEPT EXPLAINABILITY EVALUATION

Having introduced the pseudo-concept maps and their spatial characteristics, this section evaluates alignment between Grad-CAM, SHAP, and LIME explanations and the asymmetry, border-irregularity, and colour-heterogeneity reference regions. The analysis considers, first, raw alignment, second lesion-size dependence, third diagnostic-group differences, and finally chance-adjusted enrichment.

A. Raw Pseudo-Concept Alignment - Research Question 2

To what extent do XAI explanations align with clinically inspired pseudo-concept regions derived from lesion masks?

³See the supplementary appendix: Section 1.2

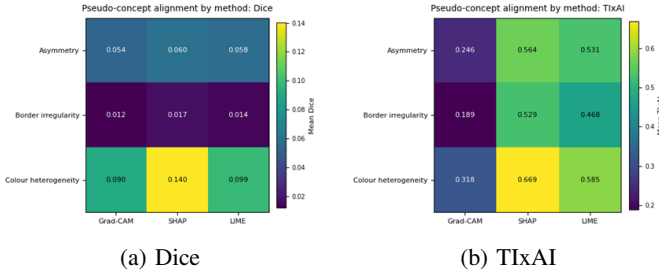


Fig. 3: Mean pseudo-concept alignment across XAI methods. Dice measures binary overlap between thresholded saliency maps and pseudo-concept regions, whereas the normalised TIxAI score quantifies the concentration of continuous saliency within each pseudo-concept.

TABLE V: Best Mean Dice Scores by Evaluation Target

Evaluation Target	Best Method	Mean Dice
Lesion Mask Baseline	SHAP	0.461
Asymmetry	SHAP	0.060
Border Irregularity	SHAP	0.017
Colour Heterogeneity	SHAP	0.140

Each XAI explanation was compared with the corresponding pseudo-concept map using the evaluation metrics defined in Section VI. Figure 3(a) shows the Dice scores, while Figure 3(b) presents the corresponding TIxAI scores. The results show that for all methods, pseudo-concept alignment was substantially weaker than whole-lesion localisation, indicating that lesion-focused saliency does not necessarily correspond to specific clinically inspired structures.

Table V shows that SHAP achieved the strongest raw alignment overall, while colour heterogeneity produced the highest pseudo-concept scores. Border irregularity remained the most challenging target, consistent with its sparse spatial extent. These results show that absolute alignment depends not only on the XAI method but also on the size and geometry of the pseudo-concept region.

The large reduction from lesion-mask overlap to pseudo-concept overlap is a central finding of the study. However, the lower concept scores should not be interpreted as direct evidence that the classifier ignores these features, because sparse target geometry and small spatial shifts can strongly affect overlap-based metrics. This is examined further in RQ2a, which considers the effect on lesion size on alignment.

B. Lesion-Size Dependence (RQ2a)

Research Question 2a: To what extent does lesion size influence pseudo-concept alignment and the reliability of XAI evaluation metrics?

Results were stratified using the predefined lesion-size classes. Raw Dice decreased markedly as lesion size decreased, particularly for asymmetry and other sparse pseudo-concepts. Tiny and very tiny lesions provide fewer pixels for concept representation, making overlap scores increasingly sensitive to small spatial displacements.

TABLE VII: Mean Dice Alignment Across Lesion-Size Tiers for Asymmetry and Distance-Weighted Border Irregularity

Size	Asym.			Border		
	G-Cam	Lime	Shap	G-CAM	Lime	Shap
Normal	0.0590	0.0632	0.0650	0.0119	0.0132	0.0158
Small	0.0104	0.0131	0.0235	0.0144	0.0201	0.0344
Tiny	0.0049	0.0064	0.0113	0.0129	0.0188	0.0309
Very Tiny	0.0031	0.0034	0.0066	0.0028	0.0027	0.0055

Table VII shows that lesion size affects the two pseudo-concepts differently. Asymmetry Dice decreases sharply as lesion size decreases across all three XAI methods. For example, SHAP falls from 0.0650 for normal lesions to 0.0066 for very tiny lesions. Border-irregularity Dice also decreases for very tiny lesions, but the pattern is not monotonic across the other size tiers. For SHAP, the score increases from 0.0158 for normal lesions to 0.0344 for small lesions, remains relatively high for tiny lesions at 0.0309, and then falls sharply to 0.0055 for very tiny lesions. This contrasting behaviour indicates that lesion-size effects depend on pseudo-concept geometry, and it becomes particularly unstable when lesions occupy very few pixels.

Additional size-stratified results for IoU, Dice, SIR, and TIxAI are provided in the supplementary Appendix ⁴. These results show that binary overlap and continuous saliency-concentration metrics can respond differently to lesion size and pseudo-concept sparsity.

C. Label-Stratified Analysis (RQ2b)

Research Question 2b: For diagnostically labelled HAM10000 images, do pseudo-concept alignments differ between melanoma and non-melanoma lesions?

Pseudo-concept metrics were also compared between melanoma and non-melanoma images using the Mann-Whitney U test with Benjamini-Hochberg FDR correction. Representative significant results are shown in **Table VI**.

Most reported comparisons remained statistically significant after FDR correction, with the largest representative effects observed for Grad-CAM. Melanoma lesions generally exhibited higher alignment for several pseudo-concepts, although the magnitude of the difference depended on the XAI method and metric. These findings answer RQ2b by indicating that diagnostic-group differences are present, but should not be interpreted independently of lesion morphology and pseudo-concept prevalence.

D. Chance-Adjusted Evaluation (RQ2c)

Research Question 2c: Do XAI methods align with pseudo-concept regions more strongly than expected by random overlap within the lesion?

To assess whether observed pseudo-concept alignments exceed those expected from random spatial overlap within the lesion, a chance-adjusted validation protocol is applied for each image, XAI method, and pseudo-concept map:

⁴See the supplementary appendix: Section 3.2

TABLE VI: Representative Differences in Pseudo-Concept Alignment Between Melanoma and Non-Melanoma Lesions

Method	Concept	Metric	Mel. Mean	Non-Mel. Mean	Effect	FDR p
Grad-CAM	Colour Heterog.	SIR	0.150	0.046	0.689	< 0.001
Grad-CAM	Border Irreg.	TlxAI _{norm}	0.360	0.162	0.586	< 0.001
Grad-CAM	Asymmetry	TlxAI _{norm}	0.442	0.213	0.516	< 0.001
LIME	Asymmetry	Dice	0.081	0.058	0.270	< 0.001
SHAP	Colour Heterog.	SIR	0.107	0.076	0.234	< 0.001

- 1) Compute the observed alignment between the XAI explanation and the target pseudo-concept mask P using the selected evaluation metric. Overlap-based metrics such as Dice operate on the binary top- $k\%$ saliency region X , whereas intensity-based metrics such as SIR operate on the continuous normalised saliency map H .
- 2) Count the number of pseudo-concept pixels K located within the binary lesion mask M .
- 3) Randomly sample exactly K pixels from within M to construct an area-matched random reference region R .
- 4) Recompute the same evaluation metric between the XAI explanation and R , preserving the metric-specific XAI representation used for the observed score.
- 5) Repeat the random sampling procedure $N_{\text{random}} = 100$ times to obtain an empirical null distribution for the metric.
- 6) Compare the observed score with the corresponding random baseline using a relative enrichment measure, where values greater than 1 indicate stronger alignment than the mean random expectation.

The randomised control zone purposely breaks shape, continuity, and spatial structure. This is appropriate here because the pseudo-concept maps themselves can be sparse, disconnected, and irregular.

Relative enrichment was defined as:

$$E_{\text{metric}} = \frac{S_{\text{observed}}}{\bar{S}_{\text{random}} + \epsilon}, \quad (7)$$

where values above 1 indicate alignment greater than the mean random expectation.

TABLE VIII: Raw Dice and Chance-Adjusted Dice Enrichment

Method	Concept	Raw Dice	Enrichment
Grad-CAM	Asymmetry	0.054	0.976
Grad-CAM	Border	0.012	0.985
Grad-CAM	Colour	0.090	1.052
LIME	Asymmetry	0.058	0.815
LIME	Border	0.014	0.784
LIME	Colour	0.099	0.987
SHAP	Asymmetry	0.060	0.635
SHAP	Border	0.018	0.553
SHAP	Colour	0.140	1.063

As presented in **Table VIII**, the adjustment substantially changed the interpretation of the raw scores. SHAP achieved the highest absolute overlap, but Grad-CAM showed the greatest mean Dice enrichment above random expectation. Colour heterogeneity remained at or above the random baseline for

all three methods, while asymmetry and border irregularity exceeded the baseline only for Grad-CAM.

The scale-stratified enrichment analysis further showed that raw Dice and relative enrichment can move in opposite directions. Although raw overlap decreased for smaller lesions, enrichment generally increased for LIME and approached the random baseline for the other methods. This suggests that part of the raw scale effect reflects geometric constraints rather than explanation quality alone.

For SHAP colour heterogeneity, melanoma lesions also showed greater chance-adjusted enrichment than non-melanoma lesions (approximately 1.18 versus 1.04 for Dice enrichment, and 1.25 versus 1.10 for SIR enrichment). This indicates stronger-than-random colour-related alignment in melanoma images, while not establishing causal dependence of the classifier on colour variation.

Overall, RQ2c shows that raw overlap and chance-adjusted enrichment capture different aspects of explanation quality. Absolute scores are strongly influenced by pseudo-concept size, lesion geometry, and saliency coverage, whereas enrichment assesses whether the observed alignment exceeds an area-matched random expectation within the lesion.

X. DISCUSSION

This research evaluated explainable AI methods at two complementary levels: lesion localisation and clinically inspired pseudo-concept alignment. Together, these analyses provide a more comprehensive understanding of explanation quality than lesion-level evaluation alone.

The lesion-level evaluation demonstrated that all three XAI methods successfully concentrated saliency within lesion boundaries, with SHAP consistently achieving the highest Iou, Dice, SIR and TlxAI. However, lesion-level localisation alone provides only a coarse assessment of explanation quality. It cannot determine whether the highlighted regions correspond to clinically meaningful morphological features. When explanations were evaluated against the proposed ABCD-inspired pseudo-concepts, alignment decreased substantially for each XAI method. This confirms the central finding of this work: accurately identifying the lesion does not necessarily imply that the explanation focuses on clinically relevant structures within the lesion. SHAP remained the strongest-performing method across most metrics. However, the reduction in overlap demonstrates that concept-level localisation is considerably more challenging than whole-lesion localisation. Amongst the investigated pseudo-concepts, colour heterogeneity consis-

tently exhibited the highest alignment, whereas border irregularity produced the lowest scores. This is consistent with the spatial characteristics of the concepts themselves. Colour heterogeneity occupies larger regions, distributed throughout the lesion. Border irregularity, only representing a thin peripheral structure, is more than four times smaller. Larger regions are more accessible to saliency-based explanations. Consequently, low border scores reflect not only poor explanation quality, but also the inherent difficulty of evaluating sparse structures. The lesion-size analysis further demonstrated that pseudo-concept evaluation depends strongly on lesion morphology. As lesion size decreases, pseudo-concept regions become increasingly small, causing all metrics, Dice, IoU... to decline sharply across XAI methods. This suggests that conventional overlap metrics become progressively less stable for small lesions, where a minor spatial shift corresponds to a large proportional difference. Therefore, comparisons across lesion sizes should be interpreted with caution, as part of the observed reduction reflects geometric constraints rather than poorer explanations. The chance-adjusted evaluation provided additional insight into these geometric effects. Although SHAP achieved the highest raw overlap scores, normalising the metrics against randomly sampled lesion regions altered the ranking of the methods. Grad-CAM exhibited the greatest enrichment above random expectation for asymmetry and border irregularity, while colour heterogeneity remained enriched for all methods. Raw overlap measures absolute agreement, whereas enrichment evaluates whether that agreement exceeds what would be expected by chance. Consequently, chance-adjusted evaluation separates genuine concept alignment from overlap arising naturally through lesion size, pseudo-concept area and saliency-map coverage. Taken together, these findings indicate that evaluating explanations solely against whole-lesion masks provides an incomplete assessment of clinical interpretability. The proposed pseudo-concept framework offers a practical intermediate representation between lesion-level localisation and expert concept annotation, enabling structured and reproducible large-scale evaluation of spatial alignment with clinically inspired features.

Overall, these results support the use of complementary lesion-level, concept-level, and chance-adjusted analyses when evaluating XAI explanations in dermoscopic image classification.

XI. LIMITATIONS AND FUTURE WORK

This study has several limitations. The pseudo-concepts are automatically derived from segmentation masks and image characteristics rather than expert dermatological annotations. In particular, colour heterogeneity should be interpreted as a heuristic proxy. Furthermore, a single classifier architecture (ResNet-50) and three saliency methods were investigated. Future work could validate the pseudo-concepts with dermatologists and extend the framework to concept-supervised or concept bottleneck models. It could also incorporate additional explainability approaches such as counterfactual explanations and investigate alternative

evaluation metrics that are less sensitive to sparse structures and lesion geometry.

XII. CONCLUSION

This research proposed DermaXplain, a framework for evaluating explainable AI methods. It used automatically generated pseudo-concept maps inspired by the dermatological ABCD rule and derived from skin lesion segmentation masks. Unlike conventional lesion-level evaluation, the proposed approach assesses whether explanations align with clinically meaningful structures representing asymmetry, border irregularity, and colour heterogeneity. The lesion-level analysis showed that SHAP produced the strongest localisation within lesion boundaries, followed by LIME and Grad-CAM. However, pseudo-concept alignment was much weaker across all methods. This demonstrated that accurate lesion localisation does not necessarily correspond to clinically meaningful explanations. Colour heterogeneity showed the strongest overall alignment with XAI explanations, but it should be interpreted as a heuristic proxy because no expert concept-level ground truth was available. Border irregularity remained the most challenging pseudo-concept, likely due to its very small spatial extent. Although generated from expert-annotated lesion masks, the border-irregularity map itself was derived algorithmically rather than clinically annotated. The study also showed that lesion size strongly influences evaluation outcomes, with very small lesions producing less stable overlap measurements. Chance-adjusted evaluation further altered the ranking of the XAI methods: although SHAP achieved the highest raw overlap scores, it exhibited the lowest mean Dice enrichment, while Grad-CAM showed the strongest enrichment above random expectation. This indicates that part of the apparent raw alignment can be explained by geometric factors such as lesion size, pseudo-concept area, and saliency-map coverage, highlighting the importance of comparing observed alignment against an appropriate random baseline.

Overall, the proposed pseudo-concept framework provides a scalable and reproducible methodology for evaluating medical image explanations without requiring expert concept annotations. Although the pseudo-concepts are heuristic rather than clinical ground truth, they offer a structured intermediate representation that extends evaluation beyond simple lesion localisation. The findings suggest that future evaluation of medical XAI systems should combine lesion-level localisation, concept-based analysis, and chance-adjusted metrics to achieve a more reliable assessment of explanation quality. Such approaches may help bridge the gap between high-performing deep learning models and the clinically meaningful explanations required for reliable decision support in dermatology.

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DECLARATION OF AI USE

The authors of this research paper, declare that the conceptual framework, study design, core implementation, and primary writing of this research paper were developed originally by us. Generative AI tools such as Gemini, Claude and ChatGPT were exclusively utilised to assist in syntax debugging during code development, model fine-tuning optimisation, and grammatical polishing of the final text. In particular, ChatGPT was used for coding assistance in LaTeX syntax (tables, mathematical formulas), and pseudo-concept generation experiments in Python code.